

NFL Draft Value Model: Identifying Undervalued Players Using Machine Learning

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1 Executive Summary

NFL teams consistently face inefficiencies in the draft, often overvaluing certain prospects while overlooking others. This project applies a data-driven approach to identify undervalued players at three high-impact offensive positions: quarterback (QB), wide receiver (WR), and running back (RB).

Using NFL Combine data and college performance statistics, we developed two models per position: (1) a classification model to estimate the probability of NFL success, and (2) a regression model to predict expected draft position. The difference between predicted performance and draft position serves as a “Moneyball” indicator of undervaluation.

Three key findings emerged:

First, incorporating college production significantly improves predictive performance. Cross-validation accuracy increased from 0.561 to 0.701 for quarterbacks (+14%) and from 0.555 to 0.703 for running backs (+15%), demonstrating that on-field performance provides substantial signal beyond combine metrics alone.

Second, the classification models exhibit higher precision than recall. For quarterbacks, precision reached 67%, while running backs and wide receivers achieved 44% and 38%, respectively. However, recall ranged from 22% to 39%, indicating that the models are more reliable at confirming high-value prospects than identifying all successful players. This makes them well-suited as filtering tools rather than exhaustive selectors.

Third, draft position predictability varies significantly by position. Quarterback draft position shows minimal predictability ($R^2 \approx 0$, MAE ≈ 62 picks), suggesting that intangible factors dominate evaluation. In contrast, running backs and wide receivers exhibit moderate predictability ($R^2 \approx 0.23$, MAE ≈ 49 –52 picks), indicating more consistent valuation patterns.

Overall, we recommend that teams place greater emphasis on college production metrics, apply caution when evaluating quarterbacks using purely quantitative models, and prioritize running backs and wide receivers when targeting undervalued players. This framework is best deployed as a decision-support tool to surface high-value prospects rather than replace scouting judgment.

2 Problem Statement

This project investigates whether publicly available NFL Combine data and college football statistics can be used to predict player success and identify systematically undervalued draft prospects. We focus on three offensive positions—quarterback (QB), wide receiver (WR), and running back (RB), due to their impact on team performance and the availability of consistent historical data.

To address this problem, we develop two complementary models for each position: a classification model that estimates the probability of NFL success, and a regression model that predicts expected draft position. By comparing predicted performance to actual draft position, we identify discrepancies that signal potential undervaluation.

The objective is to provide teams with a data-driven framework for improving draft efficiency by highlighting prospects whose projected performance exceeds their expected draft cost.

3 Datasets

3.1 Competition Data: NFL Combine and Draft

Our primary dataset was sourced from the `nflverse` project via `nfl_data_py`, containing NFL Combine measurements (8,968 players, 2000–2026) and draft records with career outcomes (6,644 players, 1980–2025).

The datasets were merged using an inner join on `pfr_id` and draft season after removing records with missing identifiers, resulting in a final sample of 5,302 players.

3.2 Additional Data: College Statistics

We augmented the dataset with college football statistics from Sports Reference (1996–2025), including passing (QB), receiving (WR), and rushing (RB) metrics collected via CSV export.

These data were merged using `cfb_id`. Approximately 688 players could not be matched and were excluded from models requiring college features.

3.3 Preprocessing

Preprocessing steps were applied to standardize features and improve data quality. Positions were grouped into 10 categories, height was converted to inches, and 255 colleges were mapped to conferences.

The dataset was restricted to players drafted in 2020 or earlier to ensure stable career outcomes (`w_av`). Combine data completeness varied, with the 40-yard dash most complete (~92%) and shuttle/cone drills least complete (~60%).

College statistics were aggregated into career and final-season features, with rate metrics recalculated from counting statistics. Special teams positions and unmapped roles (84 rows) were removed.

Dataset	Records	Year Range
NFL Combine	8,968	2000–2026
Draft Picks	6,644	1980–2025
Merged Dataset	5,302	2000–2025
College Passing	14,343 seasons	1996–2025
College Receiving	58,744 seasons	1996–2025
College Rushing	46,723 seasons	1996–2025

Table 1: Summary of data sources.

4 Data Exploration

The target variable, weighted approximate value (`w_av`), exhibits a heavily right-skewed distribution, with a mean of 18.2 and median of 10.0. This reflects the typical NFL career structure, where most players have modest outcomes and a small number of elite players drive the upper tail.

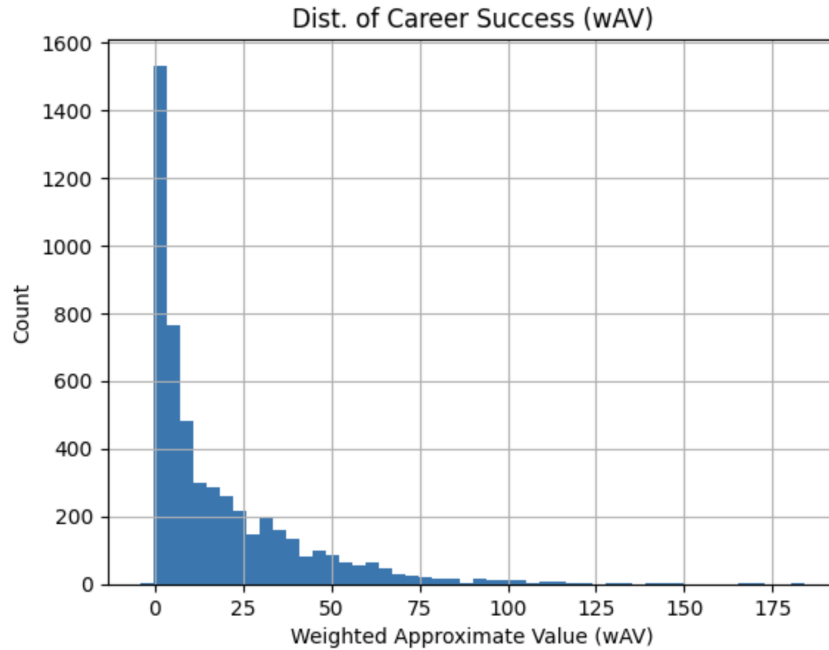


Figure 1: Distribution of career success (`w_av`).

Career outcomes vary meaningfully by position. Quarterbacks show the highest variance, indicating a high-risk, high-reward profile, while positions such as tight end exhibit more consistent but lower-ceiling outcomes. This heterogeneity suggests that predictive relationships differ by position.

Draft position is negatively associated with career success, but the relationship is highly noisy. Both high-performing late-round players and underperforming early-round selections are common, indicating inefficiencies in draft decision-making and motivating a value-based modeling approach.

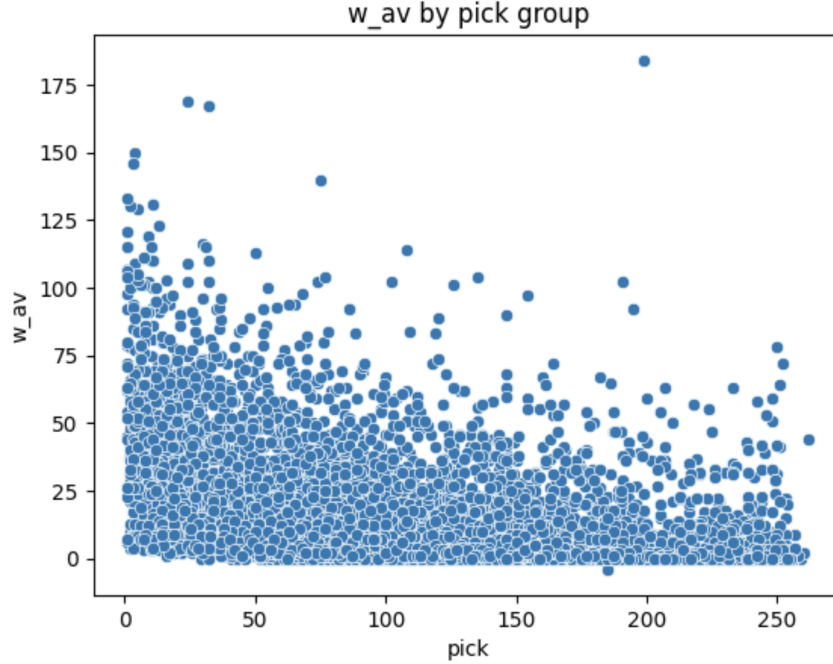


Figure 2: Draft pick vs. career success. The downward trend confirms early picks tend to perform better, but substantial noise reveals frequent misses by NFL teams.

Combine metrics show weak overall correlation with career success (generally < 0.15), but stronger position-specific relationships emerge. For example, the 40-yard dash is moderately predictive for running backs (-0.23) but near zero for wide receivers. This supports the use of position-specific models.

An ANOVA test indicates that college conference is statistically associated with career outcomes ($p = 0.003$), but incorporating conference as a feature did not improve predictive performance due to high dimensionality and sparsity. As a result, modeling efforts focus on individual-level performance metrics rather than categorical conference effects.

These findings directly inform our modeling strategy, including the use of position-specific models and the prioritization of college production features.

5 Methodology

5.1 Problem Formulation

We define NFL career success using weighted approximate value (w_{av}), a metric from Pro Football Reference that emphasizes peak performance over simple longevity. Rather than applying a single threshold across all positions, we set position-specific cutoffs by examining the distribution of w_{av} at each position and validating them against real player outcomes. Quarterbacks were held to the highest standard at $w_{av} \geq 40$ (75th percentile), capturing only multi-year starters and above. Running backs were set at $w_{av} \geq 28$ (approximately 80th percentile) and wide receivers at $w_{av} \geq 32$ (approximately 75th percentile). These thresholds were chosen so that every player classified as

a “hit” would be recognizable as a genuinely successful NFL player, not just someone who hung around a roster.

With success defined, we framed two prediction tasks for each position. The first is a binary classification problem: given a player’s pre-draft profile, will they be a hit or a bust? The second is a regression problem: given that same profile, what draft pick number should they be selected at? The gap between these two outputs is where our analysis finds value. A player the classification model rates as a likely hit but the regression model predicts will go late in the draft is, by our definition, undervalued.

5.2 Modeling Approach

We evaluated logistic regression, random forest, and gradient boosting classifiers for the hit/bust prediction, and ridge regression, lasso regression, random forest, and gradient boosting regressors for the draft pick prediction. All models were validated using 5-fold cross-validation. Gradient boosting was selected as the primary classification model at all three positions because it consistently provided the best balance of hit precision and recall. For draft pick prediction, lasso was selected for quarterbacks and running backs, while ridge was selected for wide receivers, based on cross-validated mean absolute error.

Each position was modeled independently rather than building a single universal model. Our exploratory analysis showed that the relationship between pre-draft features and career success varies significantly by position. For example, the 40-yard dash correlates meaningfully with running back success but has almost no relationship with wide receiver outcomes. A universal model would dilute these position-specific signals.

5.3 Feature Engineering and Training

Our feature set combined five NFL Combine measurables (height, weight, 40-yard dash, vertical jump, and broad jump) with position-specific college statistics. Quarterbacks received passing features such as career and final season completions, attempts, completion percentage, yards, touchdowns, interceptions, and passer rating. Wide receivers received career and final season receptions, receiving yards, yards per reception, receiving touchdowns, and yards per game. Running backs received rushing statistics along with receiving numbers to capture pass-catching ability.

Because hits represent only 20 to 35 percent of each position group, initial models tended to predict bust for nearly every player, achieving superficially high accuracy while identifying zero hits. To address this, we applied class weighting to logistic regression and random forest, and tuned the prediction threshold for gradient boosting at each position. The optimal thresholds differed by position: 0.50 for quarterbacks, 0.30 for wide receivers, and 0.20 for running backs. These reflect both the model’s confidence level at each position and the real-world risk tolerance associated with drafting at that position.

6 Modeling and Analysis

College performance provides complementary signal to combine metrics by capturing in-game production, which improves predictive performance across all models.

6.1 Hit/Bust Classification Results

Incorporating college statistics improved classification performance across all models and positions (Table 2). The largest gains occurred in logistic regression, where quarterback accuracy increased from 0.561 to 0.701 and running back accuracy from 0.555 to 0.703.

Position	Model	Combine Only CV	Combine + College CV
QB	LogReg	0.561	0.701
QB	RF	0.742	0.767
QB	GB	0.689	0.718
RB	LogReg	0.555	0.703
RB	RF	0.776	0.801
RB	GB	0.734	0.793

Table 2: Cross-validated accuracy: combine-only vs. combine + college stats.

Gradient boosting was selected as the primary model due to its superior balance of precision and recall. At optimized thresholds, precision reached 67% for quarterbacks, 44% for running backs, and 38% for wide receivers.

Threshold tuning revealed position-specific tradeoffs. Quarterbacks performed best at a threshold of 0.50, while lower thresholds improved recall for running backs (0.20) and wide receivers (0.30) without substantial precision loss (Table 3).

Position	Threshold	Hit Precision	Hit Recall	CV Accuracy
QB	0.50	67%	22%	0.718
RB	0.20	44%	39%	0.793
WR	0.30	38%	35%	—

Table 3: Hit/bust classification performance at optimized thresholds.

6.2 Draft Pick Prediction Results

Draft position predictability varies substantially by position. Quarterback draft position is largely unpredictable ($R^2 \approx 0$, $\text{MAE} \approx 62$), while running backs and wide receivers show moderate predictability ($R^2 \approx 0.23$, $\text{MAE} \approx 49\text{--}52$) (Table 4).

Position	Best Model	CV MAE (picks)	RMSE	R^2
QB	Lasso	61.6	75.2	~ 0.0
RB	Lasso	49.0	—	0.23
WR	Ridge	51.7	63.5	0.23

Table 4: Draft pick prediction performance by position.

This suggests that quarterback evaluation is driven primarily by unobserved factors, whereas running backs and wide receivers are evaluated more consistently based on measurable attributes and production.

6.3 Limitations and Uncertainties

Model recall remains moderate (22%–39%), indicating that a substantial number of successful players are not identified. Sample sizes are relatively small by position (166 QBs to 426 WRs), limiting model complexity and increasing overfitting risk.

The model does not capture key external factors such as injuries, team fit, coaching, or psychological traits, all of which contribute to outcome variability. Additionally, incomplete tracking of transfer players may understate true college production.

Finally, the absence of consistent defensive statistics prevented extension of the model to additional positions.

6.4 Areas for Further Analysis

Future work should focus on incorporating cognitive and psychological metrics (e.g., S2 testing), player grading systems (e.g., PFF), and injury history to better capture unobserved factors. Expanding the dataset to include defensive statistics and experimenting with additional models (e.g., XGBoost, neural networks) may further improve predictive performance.

As recent draft classes accumulate more data, the model can be validated and refined using out-of-sample evaluation.

7 Visualization and Interpretation

Our primary visualization is a scatter plot of predicted draft pick versus hit probability, where each point represents a prospect colored by career outcome (Figure 3). This plot captures projected performance relative to draft cost.

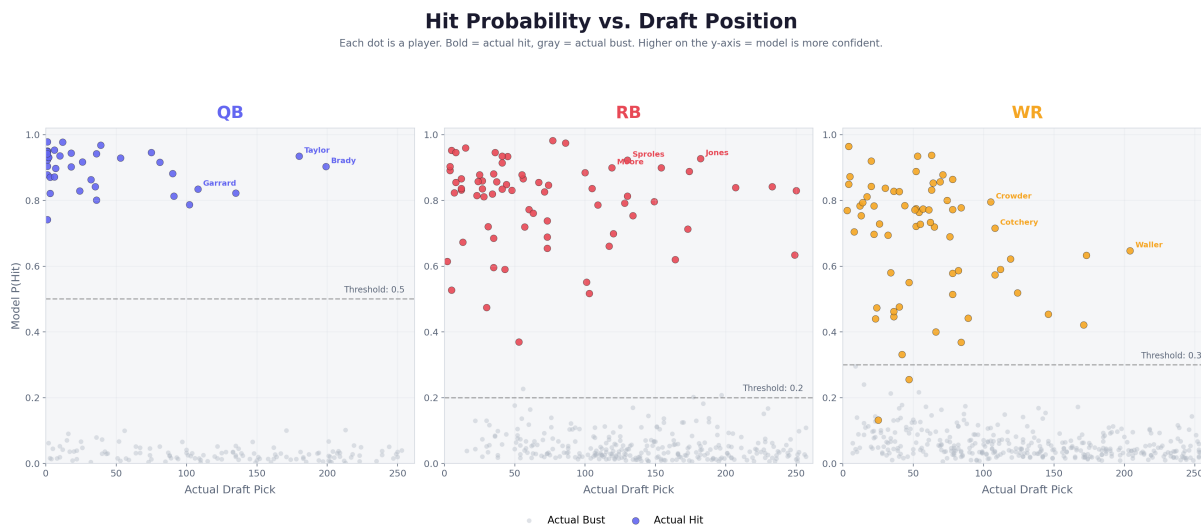


Figure 3: Hit probability vs. actual draft position. Players in the upper-right region exhibit high projected success at low draft cost.

Players in the upper-right region (Figure 4) represent undervalued prospects, where predicted performance exceeds expected draft position. Notable examples such as Tom Brady and Aaron Jones appear in this region, illustrating the model's ability to recover known draft inefficiencies.

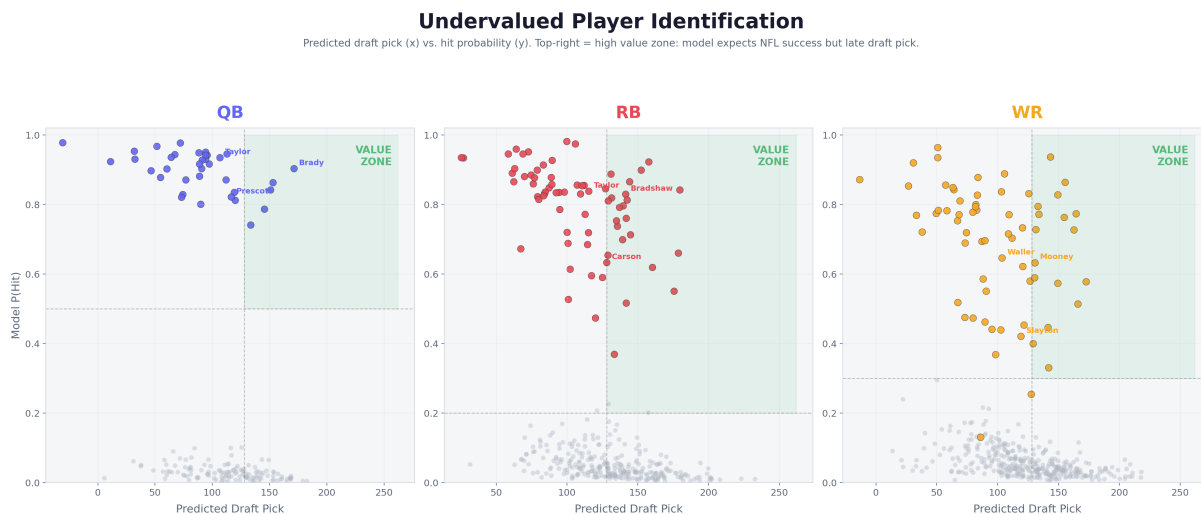


Figure 4: Undervalued player identification. The shaded region highlights high-value draft opportunities.

Draft prediction accuracy varies by position (Figure 5). Running backs and wide receivers cluster closer to the diagonal, while quarterbacks exhibit greater dispersion, reinforcing that quarterback draft outcomes are less predictable.

Points above the diagonal indicate potential overvaluation, while points below represent potential undervaluation, aligning with the model's identification of value opportunities.

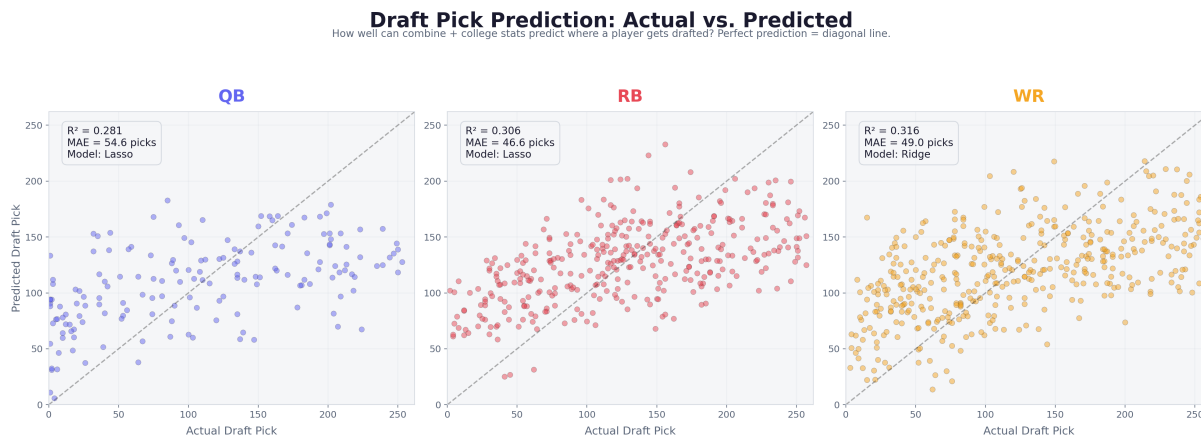


Figure 5: Actual vs. predicted draft pick. Quarterbacks show substantially greater prediction error than other positions.

8 Conclusions and Recommendations

8.1 Key Findings

Several key insights emerge from our analysis.

First, incorporating college production significantly improves predictive performance across all positions, demonstrating that on-field performance provides meaningful signal beyond combine metrics.

Second, position-specific models outperform a unified approach. Running backs and wide receivers exhibit more consistent and predictable patterns, while quarterbacks show substantially higher variability.

Third, combine measurables alone have limited predictive power and must be supplemented with performance data.

Fourth, quarterback draft outcomes are largely driven by unobserved factors. The near-zero R^2 in draft prediction indicates that intangible traits play a dominant role in evaluation.

Finally, systematic inefficiencies exist across all positions, with both late-round successes and early-round underperformance indicating that draft value is not fully optimized.

8.2 Recommendations

These findings lead to several actionable recommendations.

First, teams should prioritize college production metrics, particularly for running backs and wide receivers.

Second, model outputs should be interpreted by position. Models can be applied more aggressively for running backs and wide receivers, while quarterback evaluations require greater caution.

Third, the model should be used as a screening tool rather than a final decision-maker, as its lower recall limits its ability to identify all successful players.

Fourth, teams should invest in better measurement of intangible attributes, especially for quarterbacks, through cognitive and psychological evaluation tools.

Finally, organizations should integrate quantitative modeling with traditional scouting to improve decision-making.

8.3 Impact

This framework enables more efficient, evidence-based draft decisions by identifying players with high expected return relative to draft cost. It provides a scalable and interpretable approach for improving draft efficiency through data-driven decision support.